

Information Gathering Challenge

Name : Bharath Mourya S

Domain : Ethical Hacking

Duration : 3 months

Organization : Inlighn Tech

1. Basic Information:

Basic information about the target domain **vulnweb.com** was collected using multiple reconnaissance tools such as **WHOIS**, **nslookup**, **RED HAWK**, and **DMitry**. This phase helps in understanding domain ownership, IP address allocation, hosting details, and registration status.

Domain & WHOIS Information

The WHOIS lookup revealed the following details about the target domain:

- **Domain Name:** vulnweb.com
- **Registrar:** Gandi SAS
- **Registrar IANA ID:** 81
- **Registrar WHOIS Server:** whois.gandi.net
- **Creation Date:** 14 June 2010
- **Last Updated:** 17 November 2025
- **Expiry Date:** 14 June 2027
- **Domain Status:** clientTransferProhibited
- **DNSSEC:** Unsigned

Name Servers:

- NS-105-A.GANDI.NET
- NS-11-B.GANDI.NET
- NS-140-C.GANDI.NET

The domain is officially registered and actively maintained. The “clientTransferProhibited” status indicates protection against unauthorized domain transfers.

```
(kali㉿kali)-[~]
$ whois vulnweb.com
Domain Name: VULNWEB.COM
Registry Domain ID: 1602006391_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.gandi.net
Registrar URL: http://www.gandi.net
Updated Date: 2025-11-17T09:34:20Z
Creation Date: 2010-06-14T07:50:29Z
Registry Expiry Date: 2027-06-14T07:50:29Z
Registrar: Gandi SAS
Registrar IANA ID: 81
Registrar Abuse Contact Email: abuse@support.gandi.net
Registrar Abuse Contact Phone: +33.170377661
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Name Server: NS-105-A.GANDI.NET
Name Server: NS-11-B.GANDI.NET
Name Server: NS-140-C.GANDI.NET
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2025-12-30T13:22:19Z <<<

For more information on Whois status codes, please visit https://icann.org/epp
```

IP Address & DNS Resolution

The domain was resolved using the **nslookup** command, which successfully returned the following IP address:

- **IP Address:** 44.228.249.3

This confirms that the domain is publicly accessible and correctly mapped to its hosting server.

```
(kali㉿kali)-[~]  
$ nslookup vulnweb.com  
Server:      10.0.2.3  
Address:     10.0.2.3#53  
  
Non-authoritative answer:  
Name:   vulnweb.com  
Address: 44.228.249.3
```

Geolocation Information (RED HAWK Tool)

Geo-IP analysis was performed using the RED HAWK tool to identify the physical location of the server. The results are as follows:

- **IP Address:** 44.228.249.3
- **Country:** United States
- **State:** Oregon
- **City:** Boardman
- **Latitude:** 45.8401
- **Longitude:** -119.705

This indicates that the server hosting the website is located in a data center in Oregon, USA.

```
(kali㉿kali)-[~/RED_HAWK]  
$ php rhawk.php  
Trying Zone Transfer for vulnweb.com on ns-140-c.gandi.net ...
```

```
vulnweb.com All In One Tool For Information Gathering And Vulnerability Scanning

#####
# AEO #
#####

{C} Coded By - R3D#X0R_2H1N A.K.A Tuhinshubhra
111.aspmx.l.google.com. 121 IN A 172.217.18.26 [$] Shout Out - You ;)
112.aspmx.l.google.com. 121 IN A 172.217.18.26
113.aspmx.l.google.com. 121 IN A 142.250.101.27
114.aspmx.l.google.com. 121 IN A 192.178.164.26
[!] cURL Module Is Missing! Try 'fix' command OR Install php-curl
[!] DOM Module Is Missing! Try 'fix' command OR Install php-xml

[#] Enter The Website You Want To Scan : vulnweb.com

[#] Enter 1 For HTTP OR Enter 2 For HTTPS: 1

vulnweb.com +-----+
+-----+ List Of Scans Or Actions +-----+
+-----+

Scanning Site : http://vulnweb.com

[0] Basic Recon (Site Title, IP Address, CMS, Cloudflare Detection, Robots.txt Scanner)
[1] Whois Lookup
[2] Geo-IP Lookup
[3] Grab Banners
[4] DNS Lookup
[5] Subnet Calculator
[6] NMAP Port Scan
[7] Subdomain Scanner
[8] Reverse IP Lookup & CMS Detection
[9] SQLi Scanner (Finds Links With Parameter And Scans For Error Based SQLi)
[10] Bloggers View (Information That Bloggers Might Be Interested In)
[11] WordPress Scan (Only If The Target Site Runs On WP)
[12] Crawler
[13] MX Lookup
[A] Scan For Everything - (The Old Lame Scanner)
[F] Fix (Checks For Required Modules and Installs Missing Ones)
[U] Check For Updates
[B] Scan Another Website (Back To Site Selection)
[Q] Quit!

[#] Choose Any Scan OR Action From The Above List: 2

[+] Scanning Begins ...
[i] Scanning Site: http://vulnweb.com
[S] Scan Type : GEO-IP Lookup

[GEO-IP] IP: 44.228.249.3
[GEO-IP] Country: United States
[GEO-IP] State: Oregon
[GEO-IP] City: Boardman
[GEO-IP] Latitude: 45.8401
[GEO-IP] Longitude: -119.705

[*] Scanning Complete. Press Enter To Continue OR CTRL + C To Stop
```

Additional Information (DMitry Scan)

DMitry (Deepmagic Information Gathering Tool) was used to perform extended reconnaissance on the target domain **vulnweb.com**. This tool is designed to collect domain-related intelligence such as IP address mapping, WHOIS records, netcraft data, and potential subdomain information using passive techniques.

During the scan, DMitry successfully retrieved the following information:

- **Domain Name:** vulnweb.com
- **Resolved IP Address:** 44.228.249.3
- **Host Name Resolution:** The domain correctly resolves to the assigned IP address, confirming proper DNS configuration.
- **WHOIS Information:** DMitry confirmed the registrar as **Gandi SAS**, along with consistent domain registration and name server details matching previous WHOIS results.
- **Name Server Details:** The tool identified the same authoritative name servers associated with the domain, verifying DNS consistency.

The DMitry scan also validated that the domain is publicly accessible and properly configured for network communication. No anomalies or misconfigurations were detected during this phase.

This step helped cross-verify the accuracy of information obtained from other reconnaissance tools such as WHOIS and nslookup, ensuring reliability of the collected data.

```
(kali㉿kali)-[~]
└─$ dmitry -w vulnweb.com
Deepmagic Information Gathering Tool
"There be some deep magic going on"

HostIP:44.228.249.3
HostName:vulnweb.com

Gathered Inic-whois information for vulnweb.com
-----
Domain Name: VULNWEB.COM
Registry Domain ID: 1602006391_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.gandi.net
Registrar URL: http://www.gandi.net
Updated Date: 2025-11-17T09:34:20Z
Creation Date: 2010-06-14T07:50:29Z
Registry Expiry Date: 2027-06-14T07:50:29Z
Registrar: Gandi SAS
Registrar IANA ID: 81
Registrar Abuse Contact Email: abuse@support.gandi.net
Registrar Abuse Contact Phone: +33.170377661
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Name Server: NS-105-A.GANDI.NET
Name Server: NS-11-B.GANDI.NET
Name Server: NS-140-C.GANDI.NET
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2025-12-31T09:08:51Z <<<
```

2. Subdomain Enumeration

Subdomain enumeration was performed to identify additional subdomains associated with the target domain **vulnweb.com**. This helps in expanding the attack surface by discovering development, testing, or hidden subdomains.

The **amass enum** command was used to perform passive subdomain enumeration. Amass collected DNS records, name servers, and associated IP addresses related to the target domain.

Key findings include:

- Discovery of multiple **name servers** such as `ns-105-a.gandi.net`, `ns-11-b.gandi.net`, and `ns-140-c.gandi.net`
- Identification of **mail servers (MX records)** such as `aspmx.l.google.com` and its alternatives
- Mapping of IPv4 and IPv6 addresses related to the domain
- Identification of autonomous systems (ASNs) and netblocks associated with the hosting provider

This confirms that the domain uses Google mail services and Gandi DNS infrastructure.

```

(kali@kali)-[~]
$ amass enum -d vulnweb.com
vulnweb.com (FQDN) → ns_record → ns-105-a.gandi.net (FQDN)
vulnweb.com (FQDN) → ns_record → ns-11-b.gandi.net (FQDN)
vulnweb.com (FQDN) → ns_record → ns-140-c.gandi.net (FQDN)
vulnweb.com (FQDN) → mx_record → alt1.aspmx.l.google.com (FQDN)
vulnweb.com (FQDN) → mx_record → alt4.aspmx.l.google.com (FQDN)
vulnweb.com (FQDN) → mx_record → aspmx.l.google.com (FQDN)
vulnweb.com (FQDN) → mx_record → alt2.aspmx.l.google.com (FQDN)
vulnweb.com (FQDN) → mx_record → alt3.aspmx.l.google.com (FQDN)
alt2.aspmx.l.google.com (FQDN) → a_record → 172.217.78.26 (IPAddress)
alt2.aspmx.l.google.com (FQDN) → aaaa_record → 2607:f8b0:4001:c70::1a (IPAddress)
172.217.64.0/18 (Netblock) → contains → 172.217.78.26 (IPAddress)
15169 (ASN) → managed_by → GOOGLE - Google LLC (RIROrganization)
15169 (ASN) → announces → 172.217.64.0/18 (Netblock)
2607:f8b0:4001::/48 (Netblock) → contains → 2607:f8b0:4001:c70::1a (IPAddress)
15169 (ASN) → announces → 2607:f8b0:4001::/48 (Netblock)
vulnweb.com (FQDN) → node → rest.vulnweb.com (FQDN)
ns-105-a.gandi.net (FQDN) → a_record → 173.246.100.106 (IPAddress)
ns-105-a.gandi.net (FQDN) → aaaa_record → 2001:4b98:aaaa::6a (IPAddress)
ns-11-b.gandi.net (FQDN) → a_record → 213.167.230.12 (IPAddress)
ns-11-b.gandi.net (FQDN) → aaaa_record → 2001:4b98:aaab::c (IPAddress)
rest.vulnweb.com (FQDN) → a_record → 18.215.71.186 (IPAddress)
173.246.100.0/23 (Netblock) → contains → 173.246.100.106 (IPAddress)
2001:4b98:aaaa::/47 (Netblock) → contains → 2001:4b98:aaaa::6a (IPAddress)
2001:4b98:aaaa::/47 (Netblock) → contains → 2001:4b98:aaab::c (IPAddress)
213.167.230.0/24 (Netblock) → contains → 213.167.230.12 (IPAddress)
209453 (ASN) → managed_by → AS209453 (RIROrganization)
209453 (ASN) → announces → 173.246.100.0/23 (Netblock)
209453 (ASN) → announces → 2001:4b98:aaaa::/47 (Netblock)
209453 (ASN) → managed_by → GANDI-LIVEDNS AS for LiveDNS Anycast servers, FR (RIROrganization)
209453 (ASN) → announces → 213.167.230.0/24 (Netblock)
ns-140-c.gandi.net (FQDN) → a_record → 217.70.187.141 (IPAddress)
ns-140-c.gandi.net (FQDN) → aaaa_record → 2604:3400:aaac::8d (IPAddress)
aspmx.l.google.com (FQDN) → a_record → 74.125.130.27 (IPAddress)
aspmx.l.google.com (FQDN) → aaaa_record → 2404:6800:4003:c01::1b (IPAddress)
217.70.187.0/24 (Netblock) → contains → 217.70.187.141 (IPAddress)
2604:3400:aaac::/48 (Netblock) → contains → 2604:3400:aaac::8d (IPAddress)
18.208.0.0/13 (Netblock) → contains → 18.215.71.186 (IPAddress)
209453 (ASN) → announces → 217.70.187.0/24 (Netblock)
209453 (ASN) → announces → 2604:3400:aaac::/48 (Netblock)
14618 (ASN) → managed_by → AMAZON-AES - Amazon.com, Inc. (RIROrganization)
14618 (ASN) → announces → 18.208.0.0/13 (Netblock)
2404:6800:4003::/48 (Netblock) → contains → 2404:6800:4003:c01::1b (IPAddress)
74.125.130.0/24 (Netblock) → contains → 74.125.130.27 (IPAddress)
15169 (ASN) → announces → 2404:6800:4003::/48 (Netblock)
15169 (ASN) → announces → 74.125.130.0/24 (Netblock)
44.224.0.0/11 (Netblock) → contains → 44.228.249.3 (IPAddress)
16509 (ASN) → managed_by → AMAZON-02 - Amazon.com, Inc. (RIROrganization)
16509 (ASN) → announces → 44.224.0.0/11 (Netblock)
vulnweb.com (FQDN) → a_record → 44.228.249.3 (IPAddress)
vulnweb.com (FQDN) → node → localhost.vulnweb.com (FQDN)
localhost.vulnweb.com (FQDN) → a_record → 127.0.0.1 (IPAddress)
127.0.0.0/8 (Netblock) → contains → 127.0.0.1 (IPAddress)
0 (ASN) → managed_by → Reserved Network Address Blocks (RIROrganization)
0 (ASN) → announces → 127.0.0.0/8 (Netblock)
vulnweb.com (FQDN) → node → testphp.vulnweb.com (FQDN)
vulnweb.com (FQDN) → node → testaspnet.vulnweb.com (FQDN)
testphp.vulnweb.com (FQDN) → a_record → 44.228.249.3 (IPAddress)
testaspnet.vulnweb.com (FQDN) → a_record → 44.238.29.244 (IPAddress)
44.224.0.0/11 (Netblock) → contains → 44.238.29.244 (IPAddress)
alt3.aspmx.l.google.com (FQDN) → a_record → 173.194.76.27 (IPAddress)
alt3.aspmx.l.google.com (FQDN) → aaaa_record → 2607:f8b0:4001:c70::1b (IPAddress)
2607:f8b0:4001::/48 (Netblock) → contains → 2607:f8b0:4001:c70::1b (IPAddress)
173.194.76.0/24 (Netblock) → contains → 173.194.76.27 (IPAddress)
15169 (ASN) → announces → 173.194.76.0/24 (Netblock)
vulnweb.com (FQDN) → node → testasp.vulnweb.com (FQDN)
vulnweb.com (FQDN) → node → www.vulnweb.com (FQDN)
www.vulnweb.com (FQDN) → a_record → 44.228.249.3 (IPAddress)
alt4.aspmx.l.google.com (FQDN) → a_record → 192.178.164.27 (IPAddress)
alt4.aspmx.l.google.com (FQDN) → aaaa_record → 2607:f8b0:4004:c25::1b (IPAddress)
192.178.0.0/15 (Netblock) → contains → 192.178.164.27 (IPAddress)
2607:f8b0:4004::/48 (Netblock) → contains → 2607:f8b0:4004:c25::1b (IPAddress)
15169 (ASN) → announces → 192.178.0.0/15 (Netblock)
15169 (ASN) → announces → 2607:f8b0:4004::/48 (Netblock)
vulnweb.com (FQDN) → node → testhtml5.vulnweb.com (FQDN)
testhtml5.vulnweb.com (FQDN) → a_record → 44.228.249.3 (IPAddress)
testasp.vulnweb.com (FQDN) → a_record → 44.238.29.244 (IPAddress)

```

The enumeration has finished

The RED HAWK reconnaissance tool was launched to perform automated information gathering.

The tool provides multiple scanning options such as DNS lookup, subdomain enumeration, Geo-IP lookup, and CMS detection.

The target website **vulnweb.com** was selected for scanning, and the tool prepared the environment for subdomain discovery.

The scan menu displays multiple reconnaissance modules including:

- DNS lookup
- Subdomain scanner
- Geo-IP lookup
- Port scanning
- CMS detection

The **Subdomain Scanner** option was selected to enumerate all available subdomains associated with the target domain.

The subdomain scan successfully identified multiple subdomains related to **vulnweb.com**. Some of the discovered subdomains include:

- testphp.vulnweb.com
- testasp.vulnweb.com
- testaspnet.vulnweb.com
- testhtml5.vulnweb.com
- rest.vulnweb.com
- virus.vulnweb.com
- www.vulnweb.com
- antivirus.vulnweb.com

Each subdomain was resolved to an IP address, mostly pointing to **44.228.249.3**, confirming active subdomains hosted on the same server.

The scan completed successfully and confirmed a total of **15 subdomains** discovered for the target domain.

This demonstrates that the application has multiple environments such as testing, development, and API-related subdomains, which may increase the attack surface if not properly secured.


```
(kali@kali)-[~/RED_HAWK]
$ php rhawk.php
Trying Zone Transfer for vulnweb.com on ns-140-c.gandi.net ...
```

```
vulnweb.com All In One Tool For Information Gathering And Vulnerability Scanning

#####
RED                                     Ver 2.0.0
#####

{C} Coded By - R3D#X0R_2H1N A.K.A Tuhinshubhra
111.aspmx.l.google.com.      121      IN      A      172.217.18.26
112.aspmx.l.google.com.      121      IN      A      172.217.18.26
113.aspmx.l.google.com.      121      IN      A      143.259.101.27
114.aspmx.l.google.com.      121      IN      A      193.178.164.26
[!] cURL Module Is Missing! Try 'fix' command OR Install php-curl
[!] DOM Module Is Missing! Try 'fix' command OR Install php-xml

[#] Enter The Website You Want To Scan : vulnweb.com

[#] Enter 1 For HTTP OR Enter 2 For HTTPS: 1

#####
+-----+-----+
+                               +
+               List Of Scans Or Actions               +
+-----+-----+
#####

Scanning Site : http://vulnweb.com

8-11-b.gandi.net.      154      IN      A      217.70.187.141
[0] Basic Recon (Site Title, IP Address, CMS, Cloudflare Detection, Robots.txt Scanner)
[1] Whois Lookup
[2] Geo-IP Lookup
[3] Grab Banners
[4] DNS Lookup
[5] Subnet Calculator
[6] NMAP Port Scan
[7] Subdomain Scanner
[8] Reverse IP Lookup & CMS Detection
[9] SQLi Scanner (Finds Links With Parameter And Scans For Error Based SQLi)
[10] Bloggers View (Information That Bloggers Might Be Interested In)
[11] WordPress Scan (Only If The Target Site Runs On WP)
[12] Crawler
[13] MX Lookup
[A] Scan For Everything - (The Old Lamé Scanner)
[F] Fix (Checks For Required Modules and Installs Missing Ones)
[U] Check For Updates
[B] Scan Another Website (Back To Site Selection)
[Q] Quit!

Trying Zone Transfer for vulnweb.com on ns-140-c.gandi.net ...
SERV record query failed: NOTAUTH

[#] Choose Any Scan OR Action From The Above List: 7
```

```

[+] Scanning Begins ...
[i] Scanning Site: http://vulnweb.com 5691 IN A 44.228.249.3
[S] Scan Type : Subdomain Scanner
[i] Total Subdomains Found : 15

[+] Subdomain: Host: testphp.vulnweb.com
[-] IP: 44.228.249.3
[+] Subdomain: testphp.vulnweb.com on ns-140-c.gandi.net ...
[-] IP: 44.228.249.3
[+] Subdomain: antivirus1.vulnweb.com 154 IN A 213.173.217.7
[-] IP: 44.228.249.3 154 IN A 213.173.217.7

[+] Subdomain: www.test.php.vulnweb.com
[-] IP: 44.228.249.3

[+] Subdomain: protocoltestphp.vulnweb.com
[-] IP: 44.228.249.3
[+] Subdomain: protocoltestphp.vulnweb.com on ns-140-c.gandi.net ...
[-] IP: 44.228.249.3
[+] Subdomain: rest.vulnweb.com 121 IN A 192.172.142.1
[-] IP: 18.215.71.186 121 IN A 192.172.142.1
[+] Subdomain: testasp.vulnweb.com 121 IN A 74.125.233.100
[-] IP: 44.238.29.244

[+] Subdomain: testaspnet.vulnweb.com
[-] IP: 44.238.29.244

[+] Subdomain: testhtml5.vulnweb.com on ns-140-c.gandi.net ...
[-] IP: 44.228.249.3
[+] Subdomain: testphp.vulnweb.com on ns-105-a.gandi.net ...
[-] IP: 44.228.249.3
[+] Subdomain: testsp.vulnweb.com on ns-11-b.gandi.net ...
[-] IP: 44.228.249.3
[+] Subdomain: tetphp.vulnweb.com
[-] IP: 44.228.249.3

[+] Subdomain: virus.vulnweb.com
[-] IP: 44.228.249.3
[+] Subdomain: www.virus.vulnweb.com
[-] IP: 44.228.249.3

[+] Subdomain: viruswall.vulnweb.com
[-] IP: 44.228.249.3

[+] Subdomain: www.vulnweb.com
[-] IP: 44.228.249.3

[*] Scanning Complete. Press Enter To Continue OR CTRL + C To Stop

```

3. DNS Information

The **dnsrecon** tool was used to perform comprehensive DNS enumeration on the domain **vulnweb.com**. The scan identified multiple DNS records and confirmed that DNSSEC is not configured.

The following records were discovered:

- **SOA Record:**
 - Primary name server: ns1.gandi.net
 - Indicates the authoritative DNS server for the domain.
- **NS Records:**
 - ns-105-a.gandi.net
 - ns-11-b.gandi.net
 - ns-140-c.gandi.net
- **MX Records (Mail Servers):**
 - aspmx.l.google.com
 - alt1.aspmx.l.google.com
 - alt2.aspmx.l.google.com
 - alt3.aspmx.l.google.com
 - alt4.aspmx.l.google.com

These results confirm that Google Mail servers are used for handling email services.

- **A Record:**
 - vulnweb.com → 44.228.249.3
- **TXT Record:**
 - google-site-verification value
 - SPF record (v=spf1 -all) indicating email authentication configuration
- **SRV Records:**
 - No SRV records were found for the domain.

This confirms that the domain uses Gandi for DNS hosting and Google Workspace for email services.

```
(kali@kali)-[~]
$ dnsrecon -d vulnweb.com
[*] std: Performing General Enumeration against: vulnweb.com ...
[-] DNSSEC is not configured for vulnweb.com
[*] SOA ns1.gandi.net 173.246.100.2
[*] SOA ns1.gandi.net 2001:4b98:aaaa::2
[*] NS ns-140-c.gandi.net 217.70.187.141
[*] NS ns-140-c.gandi.net 2604:3400:aaac::8d
[*] NS ns-11-b.gandi.net 213.167.230.12
[*] NS ns-11-b.gandi.net 2001:4b98:aaab::c
[*] NS ns-105-a.gandi.net 173.246.100.106
[*] NS ns-105-a.gandi.net 2001:4b98:aaaa::6a
[*] MX alt1.aspmx.l.google.com 192.178.163.26
[*] MX aspmx.l.google.com 142.251.12.27
[*] MX alt3.aspmx.l.google.com 142.250.101.27
[*] MX alt4.aspmx.l.google.com 192.178.164.27
[*] MX alt2.aspmx.l.google.com 172.217.78.27
[*] MX alt1.aspmx.l.google.com 2607:f8b0:400e:c17::1a
[*] MX aspmx.l.google.com 2404:6800:4003:c11::1b
[*] MX alt3.aspmx.l.google.com 2607:f8b0:4023:c06::1b
[*] MX alt4.aspmx.l.google.com 2607:f8b0:4023:2009::1b
[*] MX alt2.aspmx.l.google.com 2607:f8b0:4023:1c05::1a
[*] A vulnweb.com 44.228.249.3
[*] TXT vulnweb.com google-site-verification=4LQORV-lTi-d4GPxtBEQWmFnwff7UAazQc9gZvHukbw
[*] TXT vulnweb.com v=spf1 -all
[*] Enumerating SRV Records
[-] No SRV Records Found for vulnweb.com
```

The **dig vulnweb.com mx** command was executed to retrieve mail exchange records.

The output confirmed:

- Multiple Google mail servers configured for redundancy and load balancing
- Priority-based MX records pointing to:
 - alt1.aspmx.l.google.com
 - alt2.aspmx.l.google.com
 - alt3.aspmx.l.google.com
 - alt4.aspmx.l.google.com
 - aspmx.l.google.com

Additionally, authoritative name servers were listed as:

- ns-11-b.gandi.net
- ns-105-a.gandi.net
- ns-140-c.gandi.net

This confirms that the email infrastructure is properly configured and managed using Google's mail servers.

```
(kali@kali)-[~]
$ dig vulnweb.com mx

; <<>> DiG 9.18.16-1-Debian <<>> vulnweb.com mx
;; global options: +cmd
;; Got answer:
;; -->HEADER<-- opcode: QUERY, status: NOERROR, id: 17168
;; flags: qr rd ra; QUERY: 1, ANSWER: 5, AUTHORITY: 3, ADDITIONAL: 7

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:;; udp: 4096
; COOKIE: 54fe00cb4b7c51234de95cc46954e5130bffa027a51cc797 (good)
;; QUESTION SECTION:
;vulnweb.com.                IN      MX

;; ANSWER SECTION:
vulnweb.com.                10721   IN      MX      10 alt2.aspmx.l.google.com.
vulnweb.com.                10721   IN      MX      10 alt1.aspmx.l.google.com.
vulnweb.com.                10721   IN      MX      10 alt3.aspmx.l.google.com.
vulnweb.com.                10721   IN      MX      10 alt4.aspmx.l.google.com.
vulnweb.com.                10721   IN      MX      10 aspmx.l.google.com.

;; AUTHORITY SECTION:
vulnweb.com.                167976  IN      NS      ns-11-b.gandi.net.
vulnweb.com.                167976  IN      NS      ns-140-c.gandi.net.
vulnweb.com.                167976  IN      NS      ns-105-a.gandi.net.

;; ADDITIONAL SECTION:
ns-11-b.gandi.net.         521     IN      A       213.167.230.12
ns-105-a.gandi.net.        521     IN      A       173.246.100.106
ns-140-c.gandi.net.        521     IN      A       217.70.187.141
ns-11-b.gandi.net.         521     IN      AAAA    2001:4b98:aaab::c
ns-105-a.gandi.net.        521     IN      AAAA    2001:4b98:aaaa::6a
ns-140-c.gandi.net.        521     IN      AAAA    2604:3400:aaac::8d

;; Query time: 20 msec
;; SERVER: 10.0.2.3#53(10.0.2.3) (UDP)
;; WHEN: Wed Dec 31 08:55:44 UTC 2025
;; MSG SIZE rcvd: 392
```

Zone Transfer Attempt

A zone transfer attempt was implicitly checked during DNS enumeration. The DNS servers did not allow zone transfers, which indicates that proper security controls are in place to prevent unauthorized access to DNS zone data.

4. Technology Fingerprinting

Technology fingerprinting was performed to identify the technologies, frameworks, and infrastructure used by the target website **vulnweb.com**. Multiple tools such as **WhatWeb**, **Wappalyzer**, and **BuiltWith** were used to analyze server-side and client-side technologies.

The `whatweb` tool was used to identify the web server and application details of the target website.

The results revealed the following:

- **Website URL:** `http://vulnweb.com`
- **HTTP Status:** 200 OK
- **Web Server:** nginx
- **Server Version:** nginx/1.19.0
- **IP Address:** 44.228.249.3
- **Country:** United States
- **Page Title:** *Acunetix Web Vulnerability Scanner – Test websites*

This confirms that the website is running on an **Nginx web server** and is intentionally designed as a test environment for vulnerability assessments.

```
(kali㉿Kali)-[~]  
$ whatweb vulnweb.com  
http://vulnweb.com [200 OK] Country[UNITED STATES][US], HTTPServer[nginx/1.19.0], IP  
[44.228.249.3], Title[Acunetix Web Vulnerability Scanner - Test websites], nginx[1.19.0]
```

Technology Stack Analysis (Wappalyzer)

Wappalyzer was used to detect the technologies used by the website at both frontend and backend levels.

The detected technologies include:

Programming Language

- PHP

Web Server / Reverse Proxy

- **Nginx (v1.19.0)**

Frontend Libraries

- **jQuery**

Hosting and Infrastructure

- Hosted on **Amazon Web Services (AWS)**
- Region: **United States (Oregon)**

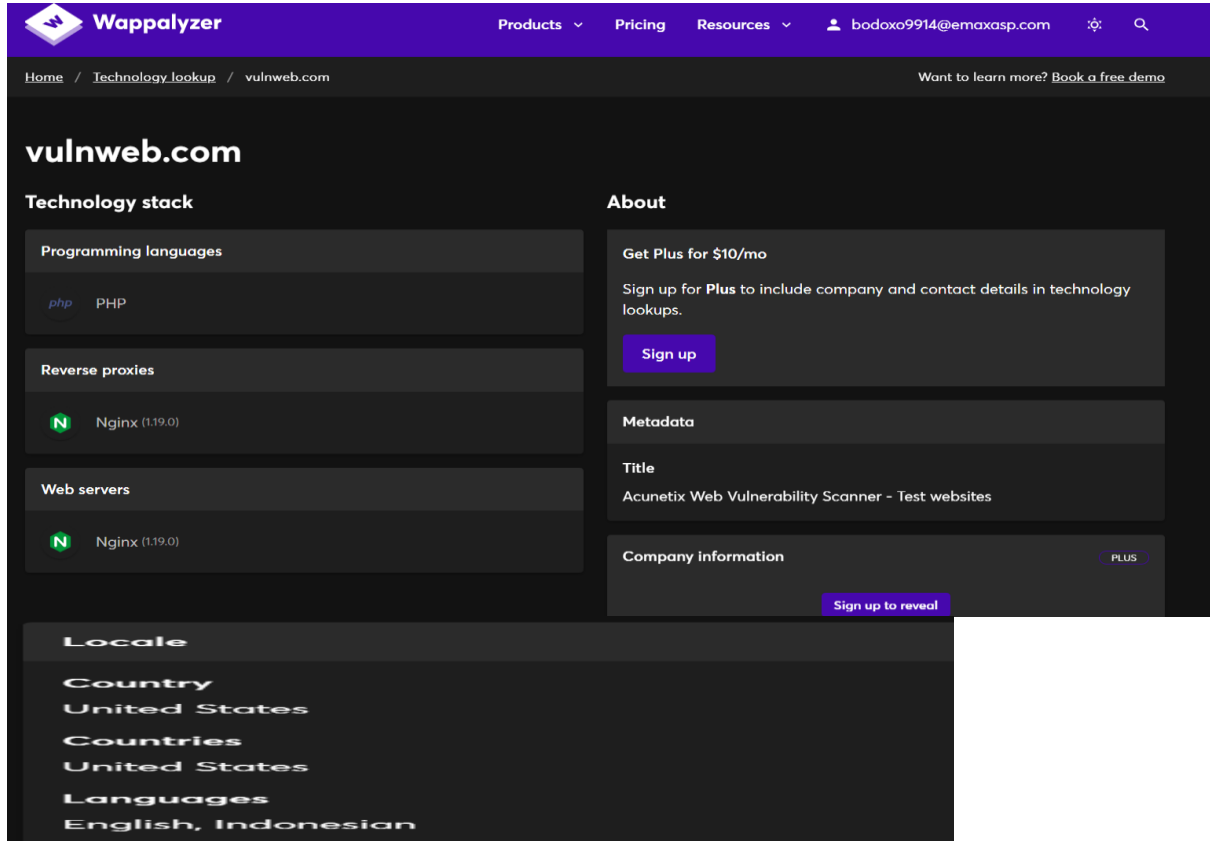
DNS and Network Services

- DNS managed by **Gandi.net**
- Reverse proxy and web server managed through **Nginx**

Localization

- Supported languages: **English, Indonesian**

This confirms that the application is hosted on cloud infrastructure using modern web technologies.



The screenshot shows the Wappalizer website interface. The top navigation bar is purple with the Wappalizer logo and links for Products, Pricing, Resources, and a user profile. The main content area is dark gray and displays the results for a technology lookup on vulnweb.com. The results are organized into sections: Technology stack, About, and Locale. The Technology stack section lists Programming languages (PHP), Reverse proxies (Nginx 1.19.0), and Web servers (Nginx 1.19.0). The About section includes a sign-up prompt for Plus (\$10/mo), Metadata (Title: Acunetix Web Vulnerability Scanner - Test websites), and Company information (Sign up to reveal). The Locale section shows Country (United States), Countries (United States), and Languages (English, Indonesian).

Wappalizer Products Pricing Resources bodoxo9914@emaxasp.com

Home / Technology lookup / vulnweb.com Want to learn more? [Book a free demo](#)

vulnweb.com

Technology stack

Programming languages

php PHP

Reverse proxies

N Nginx (1.19.0)

Web servers

N Nginx (1.19.0)

About

Get Plus for \$10/mo

Sign up for Plus to include company and contact details in technology lookups.

[Sign up](#)

Metadata

Title

Acunetix Web Vulnerability Scanner - Test websites

Company information [PLUS](#)

[Sign up to reveal](#)

Locale

Country

United States

Countries

United States

Languages

English, Indonesian

BuiltWith Technology Profiling

BuiltWith was used to analyze additional infrastructure and service dependencies used by the target website.

The following technologies were identified:

Web Hosting

- **Amazon EC2 (AWS Cloud Infrastructure)**
- Hosting region: **Amazon Oregon Region**

Content Delivery and Performance

- **Amazon S3 CDN**
- **Cloudflare Radar presence** (indicating traffic visibility and popularity)

Web Server

- **Nginx**

Programming Framework

- **PHP**

Email Services

- **Google Workspace / Google Apps for Business**
- SPF records enabled for email authentication

Analytics & Indexing

- Website listed under **Common Crawl Dataset**
- Indexed in **Chrome UX Report (CrUX)** dataset

Webmaster Tools

- Registered with **Google Webmaster Tools**

VULNWEB.COM

[Technology Profile](#)
[Detailed](#)
[Meta](#)
[Products](#)
[Performance](#)
[Relationship](#)
[Redirect](#)
[Recommendations](#)
[Company](#)

Widgets

[Global Trends](#)

CrUX Dataset

[CrUX Dataset Usage Statistics](#) · [Download List of All Websites using CrUX Dataset](#)

CrUX is a data collection system that gathers information about how real users interact with websites. This website is included in the user experiences data gathered from Google Chrome and thus considered sufficiently popular on the Internet.

CrUX Top 5m

[CrUX Top 5m Usage Statistics](#) · [Download List of All Websites using CrUX Top 5m](#)

Relative measure of site popularity within the CrUX dataset, measured by the total number of navigations on the origin. This site is in the top 5 million.

Cloudflare Radar

[Cloudflare Radar Usage Statistics](#) · [Download List of All Websites using Cloudflare Radar](#)

The website appears on the Cloudflare Radar Top 1m sites list

Cloudflare Radar Top 100k

[Cloudflare Radar Top 100k Usage Statistics](#) · [Download List of All Websites using Cloudflare Radar Top 100k](#)

The website appears in the Cloudflare Radar Top 100,000.

Common Crawl

[Common Crawl Usage Statistics](#) · [Download List of All Websites using Common Crawl](#)

This website was found in the Common Crawl dataset. Data from this site was probably used to train AI LLMs.

CommonCrawl Top 250k

[CommonCrawl Top 250k Usage Statistics](#) · [Download List of All Websites using CommonCrawl Top 250k](#)

This website appears in the Common Crawl Page Rank top 250k websites.

Frameworks

[Global Trends](#)

PHP

[PHP Usage Statistics](#) · [Download List of All Websites using PHP](#)

PHP is a widely used general-purpose scripting language that is especially suited for Web development and can be embedded into HTML.

JavaScript Libraries and Functions

[Global Trends](#)

jQuery

[jQuery Usage Statistics](#) · [Download List of All Websites using jQuery](#)

jQuery is a fast, concise, JavaScript library that simplifies how you traverse HTML documents, handle events, perform animations, and add Ajax interactions to your web pages. jQuery is designed to change the way that you write JavaScript.

[JavaScript Library](#)

Name Server

[Global Trends](#)

EuroDNS DNS

[EuroDNS DNS Usage Statistics](#) · [Download List of All Websites using EuroDNS DNS](#)

DNS services provided by EuroDNS.

gandi.net DNS

[gandi.net DNS Usage Statistics](#) · [Download List of All Websites using gandi.net DNS](#)

DNS services provided by gandi.net.

Email Hosting Providers

[Global Trends](#)

Google Apps for Business

[Google Apps for Business Usage Statistics](#) · [Download List of All Websites using Google Apps for Business](#)

Web-based email, calendar, and documents for teams. Renamed to Google Apps for Work, but now known as G Suite from Google Cloud.

[Business Email Hosting](#)

SPF

[SPF Usage Statistics](#) · [Download List of All Websites using SPF](#)

The Sender Policy Framework is an open standard specifying a technical method to prevent sender address forgery.

Web Hosting Providers
 Amazon Amazon Usage Statistics · Download List of All Websites using Amazon This site is hosted on Amazon AWS EC2 Infrastructure. Cloud Hosting Cloud PaaS
 Amazon Oregon Region Amazon Oregon Region Usage Statistics · Download List of All Websites using Amazon Oregon Region Amazon Hosted EC2 Instances in Oregon US hosting
 U.S. Server Location U.S. Server Location Usage Statistics · Download List of All Websites using U.S. Server Location The web server is located in the United States. Server Location

Web Servers
 nginx nginx Usage Statistics · Download List of All Websites using nginx nginx [engine x] is a HTTP server and mail proxy server written by Igor Sysoev.

Verified CDN
 Amazon S3 CDN Amazon S3 CDN Usage Statistics · Download List of All Websites using Amazon S3 CDN Amazon S3 is storage for the Internet. It is designed to make web-scale computing easier for developers.

Web Master Registration
 Google Webmaster Google Webmaster Usage Statistics · Download List of All Websites using Google Webmaster Webmaster tools provide you with a free and easy way to make your site more Google-friendly.

5. Directory and File Discovery

Directory and file enumeration was performed to identify hidden or accessible directories on the target website **vulnweb.com**. This process helps in discovering sensitive paths, misconfigured directories, or files that may expose information about the application.

The tool **DIRB** was used for this purpose, which performs brute-force directory scanning using a predefined wordlist.

Tool Used

- **DIRB v2.22**
- Wordlist: `/usr/share/dirb/wordlists/common.txt`

Scan Details

- **Target URL:** `http://vulnweb.com`

- **Total Words Tested:** 4612
- **Scan Start Time:** Wed Dec 31 07:32:54 2025
- **Scan End Time:** Wed Dec 31 07:52:15 2025

Discovered Directories and Files

The following directories/files were identified during the scan:

- **/alpha** → HTTP Status Code **500**
- **/index.html** → HTTP Status Code **200** (Accessible page)
- **/offerdetail** → HTTP Status Code **500**

The presence of HTTP 200 responses indicates publicly accessible resources, while HTTP 500 responses may indicate server-side errors or misconfigured endpoints.

```
(kali㉿kali)-[~]  
$ dirb http://vulnweb.com  
  
_____  
DIRB v2.22  
By The Dark Raver  
_____  
  
START_TIME: Wed Dec 31 07:32:54 2025  
URL_BASE: http://vulnweb.com/  
WORDLIST_FILES: /usr/share/dirb/wordlists/common.txt  
  
_____  
pp.html  
GENERATED WORDS: 4612  
  
—— Scanning URL: http://vulnweb.com/ ——  
+ http://vulnweb.com/alpha (CODE:500|SIZE:579)  
+ http://vulnweb.com/index.html (CODE:200|SIZE:4018)  
+ http://vulnweb.com/offerdetail (CODE:500|SIZE:579)  
  
_____  
END_TIME: Wed Dec 31 07:52:15 2025  
DOWNLOADED: 4612 - FOUND: 3
```

6. Open Ports and Services

Port scanning was performed to identify open ports and the services running on the target system **vulnweb.com**. Tools such as **Nmap**, **Masscan**, and **DMitry** were used to ensure accurate and reliable results.

Nmap Scan Results

The `nmap -sV vulnweb.com` command was used to detect open ports and identify running services.

Findings:

- **Host Status:** Up and reachable
- **IP Address:** 44.228.249.3
- **Open Port:**
 - **80/tcp** – HTTP
- **Service:** nginx
- **Version:** nginx 1.19.0

The scan confirmed that only port **80 (HTTP)** is open and actively serving web traffic. All other ports were filtered or closed, indicating a restricted and controlled network exposure.

```
(kali㉿kali)-[~]
└─$ nmap -sV vulnweb.com
Starting Nmap 7.94 ( https://nmap.org ) at 2025-12-30 21:23 UTC
Nmap scan report for vulnweb.com (44.228.249.3)
Host is up (0.26s latency).
rDNS record for 44.228.249.3: ec2-44-228-249-3.us-west-2.compute.amazonaws.com
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
80/tcp    open  http    nginx 1.19.0

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 47.31 seconds
```

Masscan Results

A high-speed scan using **Masscan** was performed across all ports (0–65535) to verify open ports.

Results:

- **Open Port Detected:** 80/tcp
- No other open ports were found during the scan.

This confirms that the server exposes only the HTTP service to the internet.

```
(kali㉿Kali)-[~]  
$ sudo masscan 44.228.249.3 --ports 0-65535  
Starting masscan 1.3.2 (http://bit.ly/14GZzcT) at 2025-12-31 10:32:34 GMT  
Initiating SYN Stealth Scan  
Scanning 1 hosts [65536 ports/host]  
Discovered open port 80/tcp on 44.228.249.3
```

DMitry Port Scan Results

The **DMitry** tool was used for additional port validation.

Findings:

- **Host:** vulnweb.com
- **IP Address:** 44.228.249.3
- **Open Port:** 80/tcp
- **Scan Summary:** 150 ports scanned, no additional open ports found.

This confirms consistency across all scanning tools.

```
(kali㉿Kali)-[~]  
$ dmitry -p vulnweb.com  
Deepmagic Information Gathering Tool  
"There be some deep magic going on"  
  
HostIP:44.228.249.3  
HostName:vulnweb.com  
  
Gathered TCP Port information for 44.228.249.3  
_____  
  
Port          State  
80/tcp        open  
  
Portscan Finished: Scanned 150 ports, 0 ports were in state closed  
  
All scans completed, exiting
```

7. Email Harvesting

Email harvesting was performed to identify publicly available email addresses and related information associated with the target domain **vulnweb.com**. This process helps in understanding publicly exposed contact points and potential communication endpoints.

The tool **theHarvester** was used to collect data from multiple public sources such as search engines, certificate transparency logs, and DNS records.

Email Harvesting Using theHarvester

The following command was executed:

```
theHarvester -d vulnweb.com -b all
```

This command initiates a comprehensive search using multiple data sources including Bing, DuckDuckGo, Yahoo, HackerTarget, DNSDumpster, and others.

Results Obtained

- **Target Domain:** vulnweb.com
- **ASN Identified:** AS16509 (Amazon)
- **Total Hosts Found:** 239
- **IP Addresses Identified:**
 - 44.228.249.3
 - 44.238.29.244
- **Interesting URLs Found (Sample):**
 - <http://testasp.vulnweb.com>
 - <http://testhtml5.vulnweb.com>
 - <http://testphp.vulnweb.com>
 - <http://testphp.vulnweb.com/login.php>
 - <http://testphp.vulnweb.com/listproducts.php>
 - <http://www.vulnweb.com>

These URLs indicate multiple application endpoints that could be used for testing or development purposes.

Email Address Discovery

- **Emails Found:** None
- No publicly indexed email addresses were discovered during the scan.

This indicates that email addresses are not openly exposed through search engines or public indexing for this domain.

Additional Findings

- Multiple subdomains were identified during the harvesting process, indicating a large and diverse infrastructure.
- The domain appears to be hosted on Amazon cloud infrastructure.
- The presence of multiple test-related subdomains confirms that the platform is designed for security testing and demonstration purposes.

```
[*] Searching Bing.  
[*] Searching Anubis.  
    Searching results.  
[*] Searching Certspotter.  
[*] Searching CRTsh.  
An exception occurred: list index out of range  
Unclosed client session  
client_session: <aiohttp.client.ClientSession object at 0x7f5f57c185d0>  
[*] Searching Dnsdumpster.  
[*] Searching Duckduckgo.  
[*] Searching Hackertarget.
```

[illegible]

```
[*] Searching Subdomainfinder99.
[*] Searching Urlscan.
[*] Searching Yahoo.
[*] Searching Brave.

[*] ASNS found: 1
AS16509 system

[*] Interesting Urls found: 13
http://testasp.vulnweb.com/
http://testhtml5.vulnweb.com/
http://testphp.vulnweb.com/
http://testphp.vulnweb.com/
http://testphp.vulnweb.com/artists.php?artist=1
http://testphp.vulnweb.com/disclaimer.php
http://testphp.vulnweb.com/listproducts.php?cat=1
http://testphp.vulnweb.com/listproducts.php?cat=3
http://testphp.vulnweb.com/login.php
http://testphp.vulnweb.com/nagiosxi/login.php
http://vulnweb.com/
http://www.vulnweb.com/
https://testphp.vulnweb.com/

[*] LinkedIn Links found: 0

[*] HOOKipmi
[*] IPs found: 2
44.228.249.3
44.238.29.244

[*] No emails found.
```

[*] Hosts found: 239

0.ca-central-1.compute.vulnweb.com
20137.vulnweb.com
2238`lj.vulnweb.com
2691`lj.vulnweb.com
3dtestphp.vulnweb.com
40d.vulnweb.com
612`ln.vulnweb.com
773.us-east-1.elb.vulnweb.com
aboutus5.vulnweb.com
adminzl.vulnweb.com
ak.vulnweb.com
aktushuk`lh.vulnweb.com
alaska.vulnweb.com
always`lg.vulnweb.com
analysis-web.vulnweb.com
analyticsg.vulnweb.com
angel.vulnweb.com
antivirus1.vulnweb.com
antivirus1.vulnweb.com:44.228.249.3
anubis.vulnweb.com
ap-northeast-13.es.vulnweb.com
ap-southeast-1.elb.vulnweb.com
api1.vulnweb.com
aquarius7.vulnweb.com
arena.vulnweb.com
aries.vulnweb.com
atalanta3.vulnweb.com
aur`ll.vulnweb.com
averell.vulnweb.com
averell7.vulnweb.com
axon.vulnweb.com
axr.vulnweb.com
b2.vulnweb.com
bach7.vulnweb.com
banka.vulnweb.com
beans.vulnweb.com
biz.vulnweb.com
biz1`lj.vulnweb.com
blue.vulnweb.com
bowsprit`lk.vulnweb.com
bragg-jacs50723.vulnweb.com
bridge.vulnweb.com
bubo`lj.vulnweb.com
carbon.vulnweb.com
careers.vulnweb.com
cdatestasp3.vulnweb.com
cdn2.vulnweb.com
ceo`lg.vulnweb.com
challenge.vulnweb.com
charlie.vulnweb.com

8. Social Media and Public Data

Social media and public data analysis was conducted to identify the online presence of the target domain **vulnweb.com**, including any publicly available information, posts, or exposed sensitive data related to the organization.

This phase focuses on identifying publicly accessible digital footprints that could potentially reveal sensitive information such as employee details, credentials, internal references, or operational insights.

Online Presence Analysis

A search was conducted across popular platforms such as:

- Google Search
- LinkedIn
- GitHub
- Twitter (X)
- Public forums and indexing services

The domain **vulnweb.com** is primarily associated with **Acunetix Web Vulnerability Scanner**, which is intentionally designed as a vulnerable testing platform for security research and learning purposes.

No official social media accounts representing a real organization were identified. Most references to the domain appear in security blogs, training platforms, and cybersecurity research materials.

Publicly Available Information

During the analysis, the following observations were made:

- The website is publicly known as a **deliberately vulnerable application** used for security testing.
- References to the domain appear in documentation, tutorials, and security training labs.
- No personal employee data or organizational profiles were found.
- No exposed credentials, passwords, API keys, or sensitive configuration data were identified.

Security Observations

- The absence of exposed credentials indicates good security hygiene.
- The domain is intentionally designed for educational and testing purposes, not for real-world business operations.
- Public references to the site are informational and do not reveal sensitive internal data.